

- 7 (a) The light-dependent stage of photosynthesis occurs within chloroplasts. In this stage, electrons are emitted from the chlorophyll *a* molecules and passed to electron acceptors.

If a redox indicator, such as DCPIP, is added to a suspension of illuminated chloroplasts, electrons will be transferred to DCPIP, causing the colour of the DCPIP to change from blue to colourless.

A student investigated the effect of the wavelength of light (colour of light) on the rate of photosynthesis.

- DCPIP was added to three colorimeter tubes, each containing a suspension of chloroplasts. The chloroplast suspensions were kept in the dark until required.
- The colorimeter tubes were each exposed to light of a different colour: red, blue or green. The intensity of light was the same for all tubes, and each was exposed to light for four minutes. All other conditions were kept the same.
- The absorbance of each chloroplast suspension was measured at one-minute intervals using a colorimeter.

The results are shown in Fig. 7.1.

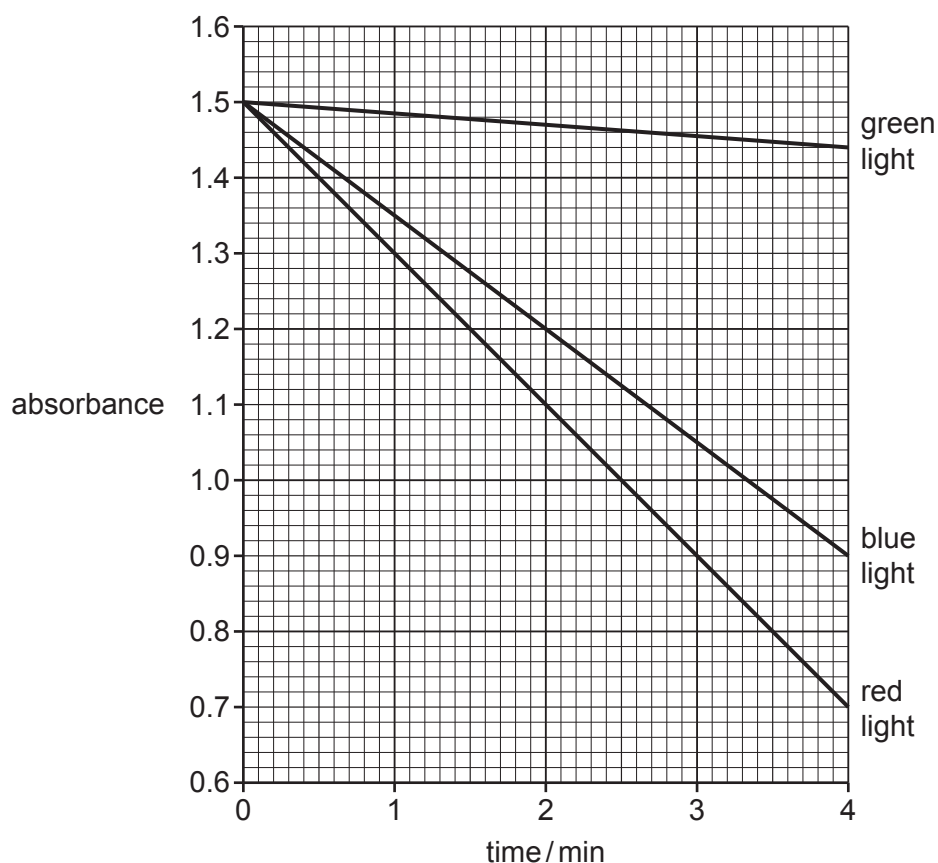


Fig. 7.1

- (i) Explain why the chloroplast suspensions were kept in the dark until required.

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..... [1]

(ii) Describe the results shown in Fig. 7.1.

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(iii) With reference to the light-dependent stage of photosynthesis, explain the differences between the results shown in Fig. 7.1 for red light and for green light.

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- (b) Changes in the atmospheric carbon dioxide concentration, light intensity and temperature can affect the rate of photosynthesis. These three factors directly affect different processes of photosynthesis.

Complete Table 7.1 using a tick (✓) to identify the processes that can be **directly** affected by each factor or a cross (✗) to identify the processes that are **not directly** affected by each factor.

Indirect effects where a change in the rate of one process affects the rate of a different process should **not** be considered.

A tick or a cross must be placed in the final column of every row.

Table 7.1

factor	process	✓ or ✗
carbon dioxide concentration	Calvin cycle	
	photophosphorylation	
light intensity	Calvin cycle	
	photophosphorylation	
temperature	Calvin cycle	
	photophosphorylation	

[3]

[Total: 10]